

The spacial profile of $\vec{B}_{\text{sec}}$ at any position $\vec{r}$ relative to the moons center is

$$\vec{B}_{\text{sec}} = \frac{\mu_0}{4\pi} \frac{3(\vec{r} \cdot \vec{m})\vec{r} - r^2\vec{m}}{r^5} \quad (\text{eqn.3})$$

where m = time dependent magnetic moment

$\vec{r}$ = vector from center to where we are observing

$$r = \sqrt{x^2 + y^2 + z^2}$$

μ_0 = vacuum permeability

$$\vec{m} = \frac{-4\pi}{\mu_0} \frac{r_m^3}{2} B_{\text{prim}} A e^{-i(\omega t - \phi)} \hat{e}_0$$

euler's formula: $e^{ix} = \cos(x) + i\sin(x)$

apply to $e^{-i(\omega t - \phi)}$

$$= \cos(\omega t - \phi) - i\sin(\omega t - \phi)$$

taking the real part: $\cos(\omega t - \phi)$

plug into $\vec{m}$

$$\vec{m} = -\frac{4\pi}{\mu_0} \frac{r_m^3}{2} B_{\text{prim}} A \cos(\omega t - \phi) \hat{e}_0$$

Now, we substitute this into eqn. (3)

$$\vec{B}_{\text{sec}} = \frac{\mu_0}{4\pi} \frac{3(\vec{r} \cdot \vec{m})\vec{r} - r^2\vec{m}}{r^5}$$

$$\vec{B}_{\text{sec}} = \frac{\mu_0}{4\pi} \frac{3 \left[\vec{r} \cdot \left(\frac{-4\pi}{\mu_0} \frac{r_m^3}{2} B_{\text{prim}} A \cos(\omega t - \phi) \hat{e}_0 \right) \vec{r} \right] - r^2 \left(\frac{-4\pi}{\mu_0} \frac{r_m^3}{2} B_{\text{prim}} A \cos(\omega t - \phi) \hat{e}_0 \right)}{r^5}$$

$$= \frac{\cancel{\mu_0}}{4\pi} \cdot \frac{\cancel{\mu_0}}{\cancel{\mu_0}} \frac{r_m^3}{2} \frac{B_{\text{prim}} A \cos(\omega t - \phi)}{r^5} [3(\vec{r} \cdot \hat{e}_0) \vec{r} - r^2 \hat{e}_0]$$

$$\vec{B}_{\text{sec}_x}(r, t) = - \frac{A B_{\text{prim}} r_m^3}{2 r^5} \cos(\omega t - \phi) [3(\vec{r} \cdot \hat{e}_{0x}) \vec{r} - r^2 \hat{e}_{0x}]$$

But, to model a rotating background, we have to

look at 2 separate physical components:

1. a primary field oscillating along the x-axis: $B_x(t) = B_{\text{prim}} \cos(\omega t)$
2. a primary field oscillating along the y-axis: $B_y(t) = B_{\text{prim}} \sin(\omega t)$ (shifted by 90° in time)

Because Maxwell's equations are linear, we solve these problems separately & add their real physical parts together at the end

linear superposition: the total response caused by multiple stimuli is the sum of the responses that would have been caused by each stimulus individually

$$\text{so, } \vec{B}_{\text{sec}_y}(r, t) = - \frac{A B_{\text{prim}} r_m^3}{2 r^5} \sin(\omega t - \phi) [3(\vec{r} \cdot \hat{e}_{0y}) \vec{r} - r^2 \hat{e}_{0y}]$$

$$\text{and } \vec{B}_{\text{sec tot}}(t) = \vec{B}_{\text{sec}_x}(t) + \vec{B}_{\text{sec}_y}(t)$$

B_{prim}

from zimmer: $\vec{B}_{\text{prim}} = B_{\text{prim}} e^{-i\omega t} \hat{e}_0$

Following same logic,

$$\vec{B}_{\text{prim}}(t) = B_{\text{prim}} \cos(\omega t) \hat{e}_{0x} + B_{\text{prim}} \sin(\omega t) \hat{e}_{0y}$$

and $\vec{B}_{\text{tot}}(t) = \vec{B}_{\text{prim}}(t) + \vec{B}_{\text{sec}}(t)$

so, the total field looks like:

$$\begin{aligned} \vec{B}_{\text{tot}}(t) = & \left[B_{\text{prim}} \cos(\omega t) \hat{e}_{0x} - \frac{AB_{\text{prim}} r_m^3}{2r^5} \cos(\omega t - \phi) [3(\vec{r} \cdot \hat{e}_{0x}) \vec{r} - r^2 \hat{e}_{0x}] \right] \\ & + \left[B_{\text{prim}} \sin(\omega t) \hat{e}_{0y} - \frac{AB_{\text{prim}} r_m^3}{2r^5} \sin(\omega t - \phi) [3(\vec{r} \cdot \hat{e}_{0y}) \vec{r} - r^2 \hat{e}_{0y}] \right] \end{aligned}$$